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Homepage

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Data Analysis

Analysis of How Genre and Release Timing Affect Steam Game Success (Positive Review Rate & User Base) (Personal Project)

GitHub: https://github.com/yehyun-42/yehyun-42.github.io/tree/main/Project/SQL_project

Tech Stack

Language	Python, SQL
IDE	Jupyter, MySQL
Data & Viz	Pandas, Matplotlib

Dataset: **Steam Market and Product Metadata**
(Source: Kaggle)

Reasons for Selecting the Dataset

Usually enjoy playing games, and aim to identify the common characteristics shared by the few titles that achieve major success among the vast array of available games.
In the case of Korean games, the dataset size is limited, and information is fragmented because each game company operates its own platform. Consequently, a Steam dataset was sourced from Kaggle, leveraging its position as a major integrated distribution platform.

Hypothesis

- Commercial success will vary depending on the game genre.
- There will be a proportional relationship between the number of users and average ratings.
- Games developed by well-known companies will have a larger user base.
- Lower prices will increase accessibility, resulting in a larger user base.
- Games released longer ago will have higher exposure, leading to a larger user base.

Data Used for Hypothesis Testing

- app_id and name. release_data. price_usd. review_score_pct. total_reviews. estimated_owners. steam_official_genres. steam_developer.

Data Preprocessing & Analysis Standard

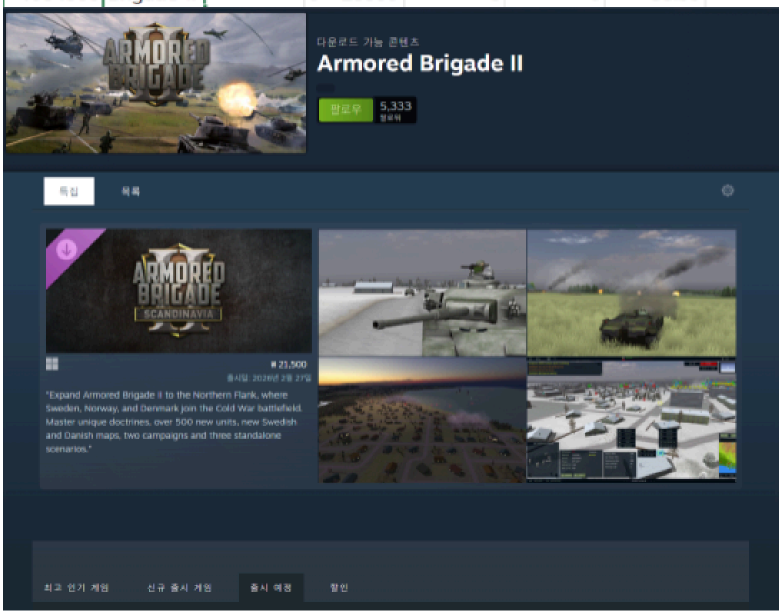
- Analysis Standard
  Positive review(review_score_pct) criteria: Steam's "Overwhelmingly Positive" standard.
  (an **average of 95% or more positive reviews**.)
- Outliers Processing Policy
  Processing: **Do not remove**
  Purpose of setting Processing Policy: The possibility that the games actually received overwhelmingly positive or negative ratings cannot be ruled out.
  Accordingly, the decision was made not to remove outliers.

Data Preprocessing

First try

- After completing preprocessing steps—such as selecting necessary columns, removing missing values, and filtering out rows containing special characters via regular expressions—Iexported the data to a CSV file using ‘to_csv’, but the DataFrame columns became misaligned.
- Attempted the preprocessing again using ‘read_excel’ instead of ‘read_csv’, but discovered a flaw within the DataFrame itself.
- To verify this, checked the DLC for ‘Armored Brigade II’ which showed a count of 10 in the “discount DLC” column, and confirmed that only one DLC has actually been released.
- It was determined that the CSV DataFrame itself is unsuitable for data analysis due to reliability issues.
- Decided to use a reliable dataframe instead.

AppID	Name	Release dá	Estimated	Peak CCU	Required á	Price	DiscountD	About the	Supported	Full audio	Reviews
2539430	Black Drag	#######	0 - 0	0	0	0	0	0		[]	[]
496350	Supipara -	29-Jul-16	0 - 20000	0	0	5.24	65	0	Springtime	['English']	[]
1034400	Mystery S	#######	0 - 20000	0	0	4.99	0	0	Immerse y	['English', '	[]
3292190	蹂꾧퉷蹂?:	#######	0 - 20000	1	0	8.99	0	1	synopsis 'I	['Korean']	['Korean']
3631080	Maze Que	#######	0 - 20000	0	0	4.99	0	0	Its not just	['English']	['English']
1654170	Agony VR	#######	0 - 20000	0	0	13.99	0	0	A JOURNE	['English', '	['English', '
1934300	Brigade II	#######	0 - 20000	8	0	35.99	10	1	Building o	['English']	[]



Discovered flaws in the DataFrame itself after completing preprocessing

Data Preprocessing

Second try

- After completing preprocessing steps—such as isolating necessary columns, removing missing values, and using regular expressions to strip special characters from titles and developer names—the data was exported as a table using 'to_csv'.
- Separated into a user table and a genre table for analysis.
- For the genre table, if a single row contains multiple genres, there is a risk that the same genre could be classified as distinct; therefore, each genre is separated into its own row.

```python
df_final['name'] = df_final['name'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_final['steam_developer'] = df_final['steam_developer'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_final.head(20)
```

	app_id	name	release_date	price_usd	review_score_pct	total_reviews	estimated_owners	steam_developer	review_score_negative
0	730	CounterStrike 2	2012-08-21	0.00	83	4980365	149410950	Valve	17
1	2868840	Slay the Spire 2	2026-03-05	24.99	97	49549	1486470	Mega Crit	3
3	3065800	Marathon	2026-03-05	39.99	90	24360	730800	Bungie	10
4	3764200	Resident Evil Requiem	2026-02-26	69.99	96	79667	2390010	CAPCOM Co Ltd	4
5	2157830	John Carpenters Toxic Commando	2026-03-12	39.99	83	2765	82950	Saber Interactive	17
6	1144200	Ready or Not	2023-12-13	49.99	85	243810	7314300	VOID Interactive	15
7	1172470	Apex Legends	2020-11-04	0.00	65	1577	47310	Respawn	35
8	1808500	ARC Raiders	2025-10-30	39.99	79	264150	7924500	Embark Studios	21
9	2852190	Monster Hunter Stories 3 Twisted Reflection	2026-03-12	69.99	75	1570	47100	CAPCOM Co Ltd	25
10	1167630	Teardown	2022-04-21	14.99	94	115777	3473310	Tuxedo Labs	6
11	2552430	KINGDOM HEARTS HD 1525 ReMIX	2024-06-13	24.99	89	8167	245010	Square Enix	11
12	4069520	Super Battle Golf	2026-02-19	7.99	95	4025	120750	Brimstone	5
14	1222670	The Sims 4	2020-06-18	0.00	79	82661	2479830	Maxis	21
15	2778610	Over The Top WWI	2026-03-06	16.26	88	3572	107160	Flying Squirrel Entertainment	12
16	359550	Tom Clancys Rainbow Six Siege	2015-12-01	0.00	75	1234092	37022760	Ubisoft Montreal	25
17	553850	HELLDIVERS 2	2024-02-08	39.99	70	807861	24235830	Arrowhead Game Studios	30
18	230410	Warframe	2013-03-25	0.00	90	2880	86400	Digital Extremes	10
19	2357570	Overwatch	2023-08-10	0.00	60	257	7710	Blizzard Entertainment Inc	40
20	236390	War Thunder	2013-08-15	0.00	76	1961	58830	Gaijin Entertainment	24
21	3472040	NBA 2K26	2025-09-04	19.99	74	12634	379020	Visual Concepts	26

```python
df_genres['name'] = df_genres['name'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_genres['steam_official_genres'] = (
    df_genres['steam_official_genres']
    .astype(str)
    .str.replace(r"[\[\]'\"]", "", regex=True))
df_genres['steam_official_genres']=df_genres['steam_official_genres'].str.split(',')
df_genres=df_genres.explode('steam_official_genres')
df_genres['steam_official_genres']=df_genres['steam_official_genres'].astype(str).str.strip()

df_genres.head(20)
```

	app_id	name	steam_official_genres
0	730	CounterStrike 2	Action
0	730	CounterStrike 2	Free To Play
1	2868840	Slay the Spire 2	Indie
1	2868840	Slay the Spire 2	Strategy
1	2868840	Slay the Spire 2	Early Access
2	3321460	Crimson Desert	Action
2	3321460	Crimson Desert	Adventure
3	3065800	Marathon	Action
4	3764200	Resident Evil Requiem	Action
4	3764200	Resident Evil Requiem	Adventure
5	2157830	John Carpenters Toxic Commando	Action
5	2157830	John Carpenters Toxic Commando	Adventure
6	1144200	Ready or Not	Action
6	1144200	Ready or Not	Adventure
6	1144200	Ready or Not	Indie
7	1172470	Apex Legends	Action
7	1172470	Apex Legends	Adventure
7	1172470	Apex Legends	Free To Play
8	1808500	ARC Raiders	Action
9	2852190	Monster Hunter Stories 3 Twisted Reflection	Adventure

Pre-processed table

Hypothesis Testing

Hypothesis 1. Commercial success will vary depending on the game genre

- Extract columns from the user ratings table(game_table.csv) and the genres table(genres_table.csv). Group by genre. Sort in descending order based on the average positive rating.

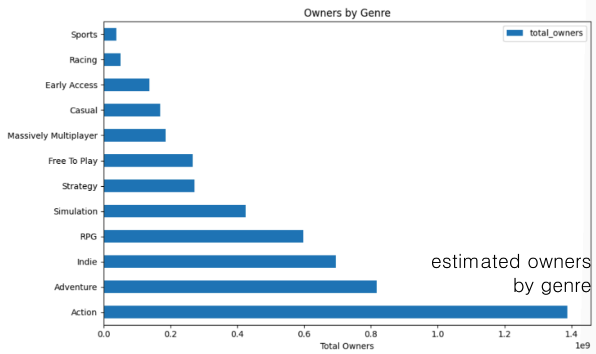
```sql
select n.steam_official_genres, avg(g.review_score_pct) as positive
from game_table g, genres_table n
where g.app_id=n.app_id
group by n.steam_official_genres
order by positive desc;
```

steam_official_genres	positive
Indie	87.1201
Casual	86.6970
Simulation	84.6833
Adventure	83.5282
Strategy	83.4787
Early Access	82.8696
Action	82.3141
Racing	81.9250
RPG	81.1877
Sports	78.3810
Massively Multiplayer	71.7969
Free To Play	69.5962

Sorted by average number of positive ratings per genre in descending order

```python
owners_by_genre.set_index('steam_official_genres')[['total_owners']].plot.barh(
    figsize=(10, 6), title='Owners by Genre', xlabel='Total Owners', ylabel=''
)

plt.tight_layout()
plt.show()
```



estimated owners by genre

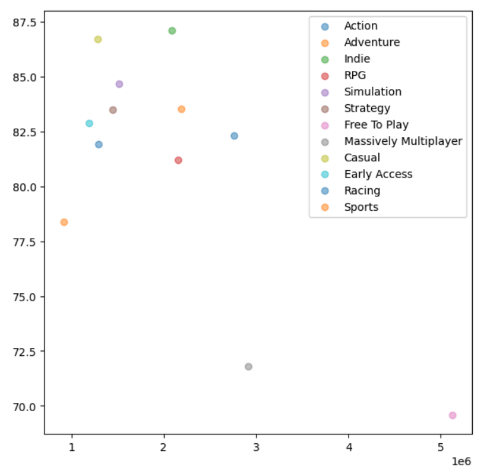
- Indie games have the highest average ratings, while free games have the lowest.
- In terms of estimated owner numbers by genre, the action genre has the highest count.
- Changes in the average value based on the estimated owner were considered; however, upon examining the lowest-rated free games and the sports genre—which has the fewest users— it was determined that the estimated owners had a negligible impact on the average.

Hypothesis 2. There will be a proportional relationship between the number of users and average ratings.

- Examine the estimated owners and the average percentage of positive reviews(review_score_pct) by genre using scatter plots.
- While free-to-play games boast the highest average estimated owners, indie games hold the highest average ratings.
- The high user count for free games appears to be driven more by price accessibility than by average ratings.

```python
fig=plt.figure(figsize=(7,7))
ax=fig.add_subplot(111)

for genres in genre_avg['steam_official_genres'].unique():
    sub=genre_avg[genre_avg['steam_official_genres']==genres]
    ax.scatter(x=sub['avg_owners'],
               y=sub['avg_positive'],
               alpha=0.5,
               label=genres)
ax.legend()
plt.show()
```



Scatter plot of estimated owners(x-axis) avg review score pct (y-axis) by genre

Hypothesis 3. Games developed by well-known companies will have a larger user base.

- Renowned game companies (such as Bethesda, Fromsoftware, and CAPCOM) tend to have high average player counts.
- However, there is also a tendency for average ratings to sometimes fall below 75%.
- It has been determined that the name of the game developer alone is sufficient to secure a certain level of user volume.
- Even when average ratings are low, the game remains within the "Positive" range on Steam, indicating relatively high satisfaction among players who purchased it based solely on the developer's reputation.

```sql
select n.steam_official_genres,
    sum(g.estimated_owners) as total_owners,
    avg(g.estimated_owners) as avg_owners
from game_table g, genres_table n
where g.app_id=n.app_id
group by n.steam_official_genres
order by total_owners desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

steam_official_genres	total_owners	avg_owners
Action	1388608890	2760653.8569
Adventure	817216260	2190928.3110
Indie	695782950	2089438.2883
RPG	597676890	2157678.3032
Simulation	425611110	1514630.2847
Strategy	272506170	1449500.9043
Free To Play	266590830	5126746.7308
Massively Multiplayer	186719040	2917485.0000
Casual	169345560	1282920.9091
Early Access	137200080	1193044.1739
Racing	51681870	1292046.7500
Sports	38629380	919747.1429

```sql
select steam_developer,
    avg(estimated_owners) as avg_owners,
    avg(review_score_pct) as avg_positive
from game_table
group by steam_developer
order by steam_developer asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

steam_developer	avg_owners	avg_positive
10 Chambers	1345200.0000	76.0000
11 bit studios	582450.0000	70.0000
24 Entertainment	5926740.0000	65.0000
343 Industries	1022280.0000	58.0000
343 Industries Splash Damag...	6985260.0000	85.0000
Acute Owl Studio	67020.0000	96.0000
AdHoc Studio	4716480.0000	95.0000
Aesir Interactive	534060.0000	73.0000
Aesir Interactive NightinGames	25470.0000	80.0000
Aether Studios	186870.0000	70.0000
Affray Interactive	352350.0000	82.0000
Afterthought LLC	2948130.0000	79.0000
[illegible]	[illegible]	[illegible]

estimated owners by genre(top left),
estimated owners by developer(top right),
estimated owners for major game companies(bottom)

Bethesda Game Studios	3658330	74.1667
FromSoftware Inc	7994620	87.8333
CAPCOM Co Ltd	2728298.571	85.2143

Hypothesis 4. Lower prices will increase accessibility, resulting in a larger user base.

- The most expensive games have user bases ranging from approximately 25,000 to 5.7 million.
- The least expensive games are all free-to-play; the user base ranges from 4,000 to over 100 million.
- It is necessary to take into account the massive success achieved by Counter-Strike 2 and PUBG: BATTLEGROUNDS.

- In the case of Destiny 2, it is important to consider that the game maintained a certain level of its existing player base after migrating to Steam following its initial success on another platform(Battle.net).
- In the case of “The Sims 4”, it is important to consider that it belongs to the “life simulation” genre and has no direct substitutes.

```sql
select name, price_usd, estimated_owners
from game_table
order by price_usd desc limit 10;
```

name	price_usd	estimated_owners
Indiana Jones and the Great...	69.99	354990
RealFlight Evolution	69.99	24390
CODE VEIN II	69.99	119790
Resident Evil Requiem	69.99	2390010
The Crew Motorfest	69.99	557670
Microsoft Flight Simulator ...	69.99	540120
WWE 2K26	69.99	25320
Sonic Racing CrossWorlds	69.99	294870
Monster Hunter Stories 3 Tw...	69.99	47100
Monster Hunter Wilds	69.99	5720670

```sql
select name, price_usd, estimated_owners
from game_table
order by price_usd asc limit 10;
```

name	price_usd	estimated_owners
CounterStrike 2	0	149410950
Fishing Planet	0	16500
PUBG BATTLEGROUNDS	0	52726470
Neverwinter	0	8880
Horizon Zero Dawn Complete ...	0	2855100
eFootball	0	36270
Apex Legends	0	47310
Call of Duty Warzone	0	293190
Crush Crush	0	16860
Dungeon Defenders II	0	89490

Top 10 most expensive games(left) and Top 10 cheapest games(right)

```sql
select name, price_usd, estimated_owners
from game_table
order by estimated_owners desc limit 10;
```

name	price_usd	estimated_owners
CounterStrike 2	0	149410950
PUBG BATTLEGROUNDS	0	52726470
Tom Clancys Rainbow Six Siege	0	37022760
Terraria	9.99	36032190
Rust	39.99	33500670
Garrys Mod	9.99	31657980
Black Myth Wukong	59.99	26084400
Stardew Valley	14.99	25384500
Cyberpunk 2077	59.99	25070400
The Witcher 3 Wild Hunt	39.99	24271650

```sql
select name, price_usd, estimated_owners
from game_table
order by estimated_owners asc limit 10;
```

name	price_usd	estimated_owners
Rec Room	0	3300
METAL GEAR SOLID MASTER COL...	59.99	3540
Solateria	17.99	3570
Wireworks	5.59	3990
VRChat	0	4440
AFTERBLAST	14.99	4770
Cargo Hunters	15.99	5040
The Lord of the Rings Online	0	5130
Rewind 99	9.59	5250
The Liar Princess and the B...	13.39	5280

Top 10 with the highest estimated owners(left)
and Top 10 with lowest estimated owners(right)

- When sorted by user count, the PvP FPS genre—rather than free-to-play games—dominates.
- It has been confirmed that the list includes not only low-priced games like ‘Terraria’, ‘Garry’s mod’, and ‘Stardew Valley’, but also high-priced titles such as ‘Rust’, ‘The Witcher 3: Wild Hunt’ and ‘Cyberpunk 2077’.
- Even among games with the smallest user bases, there is a diverse range of pricing, spanning from free to high-priced titles.
- It appears there is no direct correlation between a game’s price and its commercial success.

Hypothesis 5. Games released longer ago will have higher exposure, leading to a larger user base.

- Many of the games were released within the last year or two (2025 - 2026).
- It is also evident that some titles released in the 2010s, such as 'VRChat' and 'The Lord of the Rings Online', are included.
- This indicates that there is no direct correlation between a game's release date and its player count.

```sql
select name, release_date, estimated_owners
from game_table
order by estimated_owners asc limit 10;
```

name	release_date	estimated_owners
Rec Room	2021-09-02	3300
METAL GEAR SOLID MASTER COL...	2023-10-24	3540
Solateria	2026-03-12	3570
Wireworks	2026-03-09	3990
VRChat	2017-02-01	4440
AFTERBLAST	2025-11-26	4770
Cargo Hunters	2026-03-09	5040
The Lord of the Rings Online	2012-06-06	5130
Rewind 99	2026-03-11	5250
The Liar Princess and the B...	2026-03-11	5280

Release dates of the 10 games with the fewest players

Troubleshooting

Purpose: Selected 10 highly-rated games by genre.

Trouble: It was deemed inefficient to execute SQL statements for each of the 11 genres individually to extract data by genre.

Solution: Explore methods for simultaneous use of all genres.

First try: Using DENSE_RANK()

- Genre names are sorted in descending alphabetical order, which was not the intended result.

Second try: Using ROW_NUMBER()

- Instead of the top 10 items per genre, the top 10 items across all genres are returned.
- Use 'PARTITION BY' when want to calculate the rank of a row within a specific group based on the search results.

Third try: Using Sub Query

- Initially used 'RANK()'; while genre-based rankings were generated, duplicate ranks were not excluded, sometimes resulting in more than 10 output rows.
- Successfully limited the output to the top 10 by using 'ROW_NUMBER()'.

```sql
select g.name, n.steam_official_genres, g.review_score_pct
from game_table g, genres_table n
where g.app_id=n.app_id and n.steam_official_genres='Indie'
order by g.review_score_pct desc limit 10;
```

name	steam_official_genres	review_score_pct
The Jackbox Party Pack 3	Indie	100
Creature Kitchen	Indie	99
Opus Magnum	Indie	99
Funi Raccoon Game	Indie	99
Sam Max Beyond Time and Space	Indie	99
Crab Champions	Indie	99
Skate Story	Indie	98
Stardew Valley	Indie	98
Scarlet Hollow	Indie	98
Slime Rancher	Indie	98

Extracting the top 10 indie games

```sql
select g.name, n.steam_official_genres,
    dense_rank() over(order by n.steam_official_genres desc) as genres_rank,
    g.review_score_pct
from game_table g, genres_table n
where g.app_id=n.app_id;
```

name	steam_official_genres	genres_rank	review_score_pct
New Cycle	Strategy	1	74
Songs of Syx	Strategy	1	90
Pluto	Strategy	1	95
Slay the Spire 2	Strategy	1	97
Bloodgrounds	Strategy	1	81
Wartales	Strategy	1	67
Foundation	Strategy	1	90
Cult of the Lamb	Strategy	1	97
Machine Mind	Strategy	1	71
MENACE	Strategy	1	92

Result of using DENSE_RANK()

```sql
select g.name, n.steam_official_genres,
    row_number() over(order by review_score_pct desc) as genres_rank
from game_table g, genres_table n limit 10;
```

name	steam_official_genres	genres_rank
STAR WARS Empire at War Go...	Action	1
The Jackbox Party Pack 3	Action	2
STAR WARS Empire at War Go...	Adventure	3
The Jackbox Party Pack 3	Adventure	4
STAR WARS Empire at War Go...	Indie	5
The Jackbox Party Pack 3	Indie	6
The Jackbox Party Pack 3	Racing	7
STAR WARS Empire at War Go...	Simulation	8
The Jackbox Party Pack 3	Simulation	9
STAR WARS Empire at War Go...	Adventure	10

Result of using ROW_NUMBER()

```sql
where rankrow.genres_rank <= 10
order by rankrow.steam_official_genres asc, rankrow.genres_rank asc;
```

name	steam_official_genres	genres_rank
Funi Raccoon Game	Action	1
Crab Champions	Action	2
VTOL VR	Action	3
Portal	Action	4
Crime Scene Cleaner	Action	5
Slime Rancher	Action	6
Skate Story	Action	7
Portal 2	Action	8
Resident Evil 2	Action	9
Wobbly Life	Action	10
Funi Raccoon Game	Adventure	1
Sam Max Beyond T...	Adventure	2
Crab Champions	Adventure	3
Subnautica	Adventure	4
Skate Story	Adventure	5
Berry Bury Berry	Adventure	6
Portal 2	Adventure	7
Slime Rancher	Adventure	8
Look Outside	Adventure	9

Display the top 10 rated games by genre.

Data Analysis Results

Game success by genre

- Distinct differences in estimated owners and average ratings were observed depending on the game genre.
- The **action genre** recorded the highest figures for user numbers, while the **indie genre** showed the highest average ratings.

Proportional relationship between the estimated owners and average ratings

- **Indie games** have the highest average ratings, while **free-to-play games** have the highest average player counts.
- However, free-to-play games have the lowest average ratings.
- Rather than indicating that these games failed to meet expectations, this likely reflects their high accessibility; **a wide range of users often tries them out casually, regardless of their personal preferences.**

estimated owners according to developer awareness

- **Developers with high brand** recognition tend to **have a large average user base**.
- However, the fact that average ratings sometimes fall below 75% suggests that a significant number of purchases are **driven solely by the developer's name.**
- Nevertheless, since these titles generally maintain a "Positive" rating on Steam, it can be concluded that purchases based solely on the **developer's name still result in relatively high levels of satisfaction.**

estimated owners according to price

- Games with both large and small player bases **exhibit a diverse range of price** points.
- It appears **that price does not significantly influence a game's success.**

estimated owners according to release date

- Sorting the games by user count in ascending order and examining their release dates revealed that the list includes not only recently released titles but also games launched over a decade ago.
- While the release date appears to have some influence, **it does not seem to be a strictly proportional factor.**

Conclusion: Analyzed user density and rating distribution on the Steam platform, successfully deriving insights to determine target genres and price points for new game launches.

Project Retrospective and Areas for Improvement

1. **DataFrame Selection**

- Failed to identify issues **inherent in the CSV** file beforehand.
- Realized the necessity of conducting a thorough **preliminary check when selecting the DataFrame.**
- Recognized the need for **data cleaning techniques** to handle situations where the DataFrame cannot be modified.

2. **Lack of proficiency in grammar and analytical methods during the analysis**

- Since a single game can span multiple genres, **duplicate game titles occurred.**
- Realized too late that I should have implemented **grouping via 'GROUP BY'** or **prevented duplicates using 'DISTINCT'**.

Data Engineering

Animal Shelter Monitoring system (Team Project)

GitHub: https://github.com/yehyun-42/yehyun-42.github.io/tree/main/Project/Team_Project

Tech Stack

Language	Python
IDE	VS Code
Database	SQLAlchemy, Flask-Migrate
Web streaming server	Flask

Problem Identification

While volunteering at a **private shelter for abandoned dogs**, experienced firsthand the staffing shortages faced by a very small team managing a large number of animals.
Began seeking technological solutions to address the various problems caused by this lack of manpower.

Technical Motivation

Observed **deep learning-based object detection technology currently in practical use** at the Gwacheon Korea National Science Museum.
Aimed to build a system that alleviates human workload by applying the technology learned in the classroom to real-world scenarios.

Reasons for Selection the Topic

- Establishing protocols to handle unexpected incidents and crises arising from staff shortages in animal shelters, where a small team manages a large number of animals.
- Aiming for more efficient management and reduced human workload by utilizing drones and AI.

Benchmarking Reference

- **Technology for Estimating Penguin Populations by the Korea Sejong Station Wintering Research Team.**
  A technology that uses drones to capture footage of penguin habitats and employs AI to identify penguin populations and nests. This serves as a reference for developing capabilities to detect the populations of protected animal species.

- **Smart roadkill prevention systems installed in areas such as Korea Yangpyeong-gun, Gyeonggi-do, and Pyeongchang-gun, Gangwon-do.**
  A system designed to prevent wildlife roadkill on national highways. It utilizes AI-based CCTV and radar sensors to detect the presence of wildlife. This information serves as a reference for implementing wildlife detection capabilities.
- **Real-time mobile push notifications from the home camera.**
  A home camera function that monitors a designated area and sends an alert to the user's smartphone (or similar device) when specific conditions are met; useful for reference when setting up a real-time notification system.

Role in Project: Backend and DB pipeline development

- *Database & Data Management*
  Database creation and visualization
  Integration and Management of Detection Log Database
- Streaming Troubleshooting Fix
  Resolving ESP32-CAM Streaming Errors
- Alert Notification Function
  E-mail Alerts: E-mail integration
  Discord Alerts: Webhook integration

Data Flow

Threshold Web Dashboard → Detection Threshold DB → Threshold Dictionary → YOLO (Threshold-based Detection) → Detection History DB → Detection History Web Dashboard

Database & Data Management

Initial Concept Table

User	Users
Detected History of Protected Species	Animals
Detection History of Wild animal	Dangers
Alert Sent	Alarms

Significant

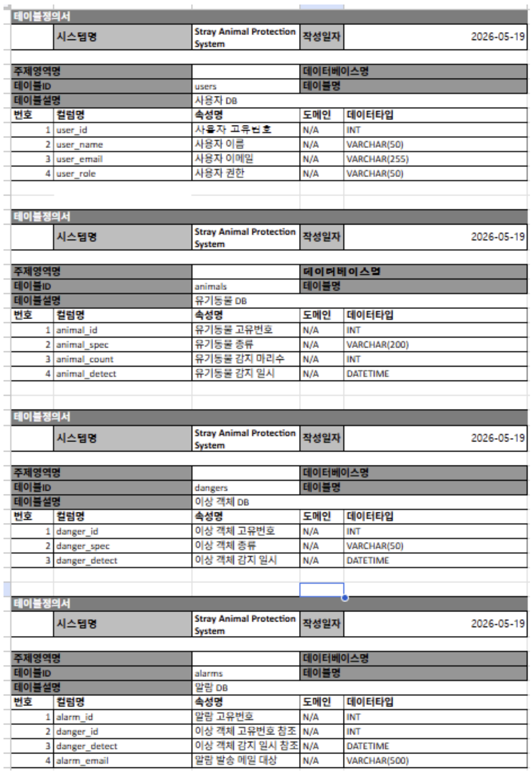
- For the alarm generation database, it is necessary to reference records from other databases.
- Reference the unique ID and detection timestamp from the wild animal detection database(Danger DB).
- Reference the email address from the User database.

Final confirmed Table

User	Users
Protected Species count	Animals
Wild animal that need Detection	Dangers
Detection History	Detect_animal

Reason for table modification

- Identified the need to track the population count(baseline) of sheltered stray animals.
- To improve efficiency, modified the system to evaluate detection data in real-time and determine whether to trigger an alert, rather than relying on a separate, tactically managed alert dispatch table.



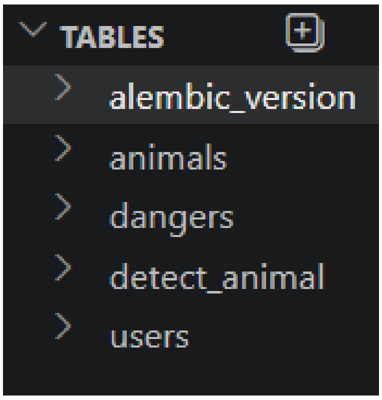
테이블정의서				
	시스템명	Stray Animal Protection System	작성일자	2026-05-19
주제영역명			데이터베이스명	
테이블ID		users	테이블명	
테이블설명		사용자 DB		
번호	컬럼명	속성명	도메인	데이터타입
1	user_id	사용자 고유번호	N/A	INT
2	user_name	사용자 이름	N/A	VARCHAR(50)
3	user_email	사용자 이메일	N/A	VARCHAR(255)
4	user_role	사용자 권한	N/A	VARCHAR(50)

테이블정의서				
	시스템명	Stray Animal Protection System	작성일자	2026-05-19
주제영역명			데이터베이스명	
테이블ID		animals	테이블명	
테이블설명		유기동물 DB		
번호	컬럼명	속성명	도메인	데이터타입
1	animal_id	유기동물 고유번호	N/A	INT
2	animal_spec	유기동물 종류	N/A	VARCHAR(200)
3	animal_count	유기동물 감지 마리수	N/A	INT
4	animal_detect	유기동물 감지 일시	N/A	DATETIME

테이블정의서				
	시스템명	Stray Animal Protection System	작성일자	2026-05-19
주제영역명			데이터베이스명	
테이블ID		dangers	테이블명	
테이블설명		이상 객체 DB		
번호	컬럼명	속성명	도메인	데이터타입
1	danger_id	이상 객체 고유번호	N/A	INT
2	danger_spec	이상 객체 종류	N/A	VARCHAR(50)
3	danger_detect	이상 객체 감지 일시	N/A	DATETIME

테이블정의서				
	시스템명	Stray Animal Protection System	작성일자	2026-05-19
주제영역명			데이터베이스명	
테이블ID		alarms	테이블명	
테이블설명		알람 DB		
번호	컬럼명	속성명	도메인	데이터타입
1	alarm_id	알람 고유번호	N/A	INT
2	danger_id	이상 객체 고유번호 참조	N/A	INT
3	danger_detect	이상 객체 감지 일시 참조	N/A	DATETIME
4	alarm_email	알람 발송 메일 대상	N/A	VARCHAR(500)

Table Definition Document



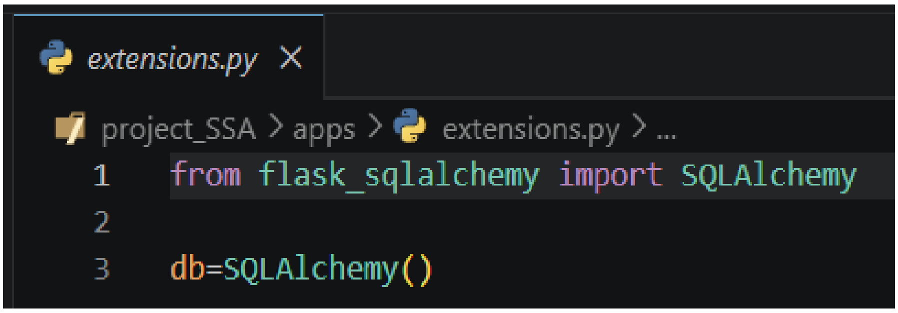
Final confirmed DB

Visualization of boolean values

- Configure the system to send an alert when specific conditions (e.g., population falling below the threshold) are met.
  Added a 'detect_alarm' column with a boolean value to visually indicate whether an alert has been triggered.
  Enabled status monitoring—specifically "Not Sent" vs "Sent"—within the visualization dashboard.

Preventing Circular Reference

Initial DB configuration structure: importing the database within the 'app.py' file.
Identified potential risks of circular dependencies and maintenance issues as the monitoring system scales.



extensions.py for Preventing Circular Reference

Solution: Create an extensions.py file and set it to be responsible only for the DB. Afterwards, import extensions.py rather than DB in app.py.

Database Visualization Dashboard

Visualize the database to facilitate easy access and management.
Visualize the stored data by applying filtering and processing, enabling users to intuitively understand the information.

The 'Animals' table is integrated to allow for immediate updates within the visualization dashboard.

- The protected animal (key) and its population count (value) are defined as a dictionary.
- The system is configured so that user inputs are applied to the dictionary.

Issues encountered during table integration

- Unable to link YOLO detection records and the 'Detect_animal' table with the threshold dictionary and the 'Animals' table.
  Cause: Tensor values and dictionary values were not converted into a format compatible with the database table.
  Solution: Directly extract the Tensor values and dictionary column values, and insert them directly into SQLite.

Retrospective

Reflection and Improvement

1. **Population Detection System**
   Shelters typically house dozens to hundreds of animals.
   Current detection logic: Data is saved to the database and an alarm is triggered only if the "below-threshold" state persists for 10 seconds after initial detection.
   This approach is difficult to implement in actual shelter environments; a more reliable method of object recognition is required.
   Techniques such as partial labeling could potentially resolve issues related to object overlapping.
2. **Wild Animal Detection System**
   Current detection logic: Detects only previously registered wildlife.
   Wildlife may change habitats for various reasons.
   The appearance of new wildlife in the area entails the inconvenience of requiring a learning and registration process.
   Logic of the benchmarked smart roadkill prevention system: Determines whether a detected moving object is wildlife.
   The logic needs to be modified to detect "wildlife" itself, rather than just "registered wildlife", by utilizing this approach.
3. **Map Service**
   Current detection logic: An alarm is triggered when wildlife appear.
   To devise a fundamental solution, a map recording Wildlife sighting locations is provided.
   It is necessary to identify key wildlife hotspots and establish a means to share information on how to deal with wildlife.

Growth

Experience in database development and integration

- Designed a data pipeline connecting the ESP32-CAM, Flask server, YOLO, and the database.
- Designed and implemented database tables tailored to specific requirements.
- Gained experience in the server-side process of storing and visualizing YOLO detection data, as well as the process of storing user-inputted information and linking it with reference thresholds.

Troubleshooting and Prevention

- Encountered and resolved unexpected software-related issues.
- Prevented potential future errors and enhanced security.

Communication between team members

- Instead of grappling with project-related issues in isolation, determine the direction and resolve them through communication with the entire team.

Data Analysis

Analyzed correlations between daily high-ozone event counts and peak heatwave/wildfire periods (Personal Project)

GitHub: https://github.com/yehyun-42/yehyun-42.github.io/tree/main/Project/Data_analysis

Tech Stack

Language	Python
IDE	Jupyter
Data & Viz	Pandas, NumPy, Matplotlib, Seaborn

Dataset: **Western North American FLEXPART Back Trajectory 1994-2021 Merge Data**
(Source: NASA)

Hypothesis

- Periods of high ozone concentrations would coincide with periods of heatwaves and wildfires.
- Since heatwaves are influenced by global warming, higher ozone concentrations are expected to be observed given the current rise in global temperatures.

Data Used for Hypothesis Testing

- Ozone levels above the standard threshold, by year, month, and day.
- Records of heatwaves and wildfires by month and day with the highest frequency of high-concentration ozone level detections.

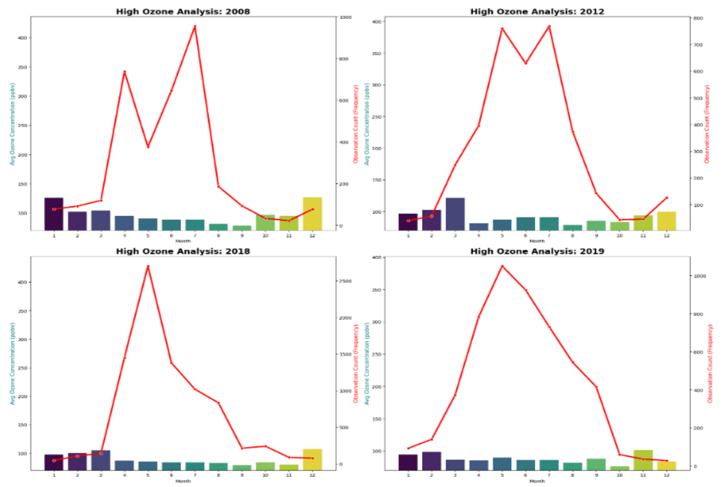
Data Preprocessing & Analysis Standard

- Analysis Standard
  Standard value: **70ppbv** (Criteria for Issuing Ozone Advisories in the U.S.)
  Purpose of setting Standard value: Since the subject of analysis is 'high-concentration ozone', this criterion was adopted to isolate the necessary data into a separate pivot table for analysis.
- Outliers Processing Policy
  Processing: **Do not remove**
  Purpose of setting Processing Policy: Since high ozone levels observed during heatwaves and wildfires were selected for analysis, values that could be classified as outliers were also included in the analysis.

Hypothesis Testing

Hypothesis 1. Periods of high ozone concentrations would coincide with periods of heatwaves and wildfires.

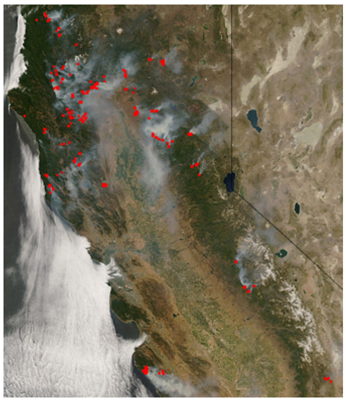
Visualization of monthly data for the years with the highest number of recorded high-concentration ozone levels (2008, 2012, 2018, and 2019).
- Sharp increase in the frequency of high-concentration ozone observations during summer(June-August).
- Heatwaves are predicted to be the cause during July and August. The months with the highest observed concentrations were selected based on data from June and September.
Records of heatwaves and wildfires for those months were then examined.



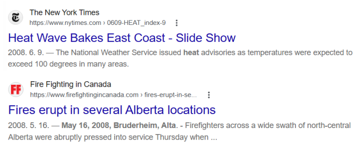
Monthly average levels of high ozone concentrations(bar) and observation conts (line) for the year with the highest number of observations.
Y-axis: Average ozone concentration (bars), number of ozone concentration observations(line)

In 2008, a record-breaking heatwave occurred in California; a massive wildfire had broken out in Alberta two months prior.
In 2018, a fire of a magnitude visible from space broke out in California.

Demonstrated a direct correlation between wildfires, heatwaves, and ozone concentrations.



Satellite image of the California wildfires released by NASA MODIS



The New York Times
https://www.nytimes.com › 0609-HEAT_index-9

Heat Wave Bakes East Coast - Slide Show

2008. 6. 9. — The National Weather Service issued **heat** advisories as temperatures were expected to exceed 100 degrees in many areas.

Fire Fighting in Canada
https://www.firefightingincanada.com › fires-erupt-in-se...

Fires erupt in several Alberta locations

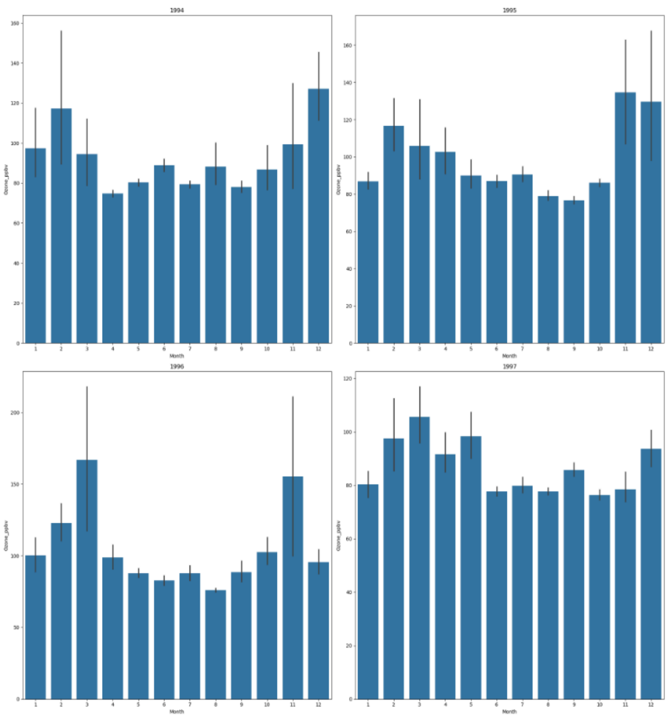
2008. 5. 16. — **May 16, 2008, Bruderheim, Alta. -** Firefighters across a wide swath of north-central Alberta were abruptly pressed into service Thursday when ...

News on the 2008 Heatwave and Fires

Hypothesis Testing

Hypothesis 2. Since heatwaves are influenced by global warming, higher ozone concentrations are expected to be observed given the current rise in global temperatures.

To compare past and present ozone concentrations, the monthly average values for high ozone concentrations were examined for the years (1994-1997) that recorded the fewest instances of ozone levels exceeding the standard.
- The average value is generally 100 ppbv; as expected, this is lower than current levels.
- However, there are cases where the number of observations increases significantly.



Average monthly high−concentration ozone levels from 1994 to 1997
Y−axis: average ozone concentration
X−axis: month

Review of annual wildfire occurrence statistics extracted using data from the National Interagency Fire Center, a U.S. wildfire statistics website.
- The number of wildfires that occurred in the past was similar to the present.
- The scale of the damage is significantly smaller compared to the present.
- Depending on the scale of the damage, the volume of air pollutants generated and the area affected by the heat from the wildfire increase.

Year	Fires	Acres
2019	50,477	4,664,364
2018	58,083	8,767,492
2012	67,774	9,326,238
2008	78,979	5,292,468
1997	66,196	2,856,959
1996	96,363	6,065,998
1995	82,234	1,840,546
1994	79,107	4,073,579

Not only the frequency of wildfires but also their scale affects ozone concentrations. Additionally, the United States announced amendments to the Clean Air Act in 1990.

Data Analysis Results

Confirmation of a direct correlation between wildfires, heatwaves, and ozone concentrations.

- It was confirmed that the summer period (June - August) coincides with the wildfire season and the period when high ozone concentrations are frequently observed.

Higher ozone levels observed compared to the past.

- When comparing the past and present, it was confirmed that ozone concentration has increased significantly.
- It was confirmed that the frequency of wildfires remains unchanged compared to the past, or was higher in the past.

Results

1. **Early response to natural disasters**
   Not only the frequency of wildfires but also their scale affects ozone formation.
   It is necessary to prevent wildfires through measures such as early suppression.
   There is a need to actively adopt air pollutant control equipment, such as nitrogen oxide reduction systems.
2. **Legal regulations and social efforts required**
   Ozone concentrations dropped to low levels in the mid-1990s (specifically 1994), shortly after the 1990 amendment to the Air Pollution Prevention Act.
   This demonstrates that ozone concentrations can be reduced through the efforts of citizens and businesses.

Project Retrospective and Areas for Improvement

1. Necessity of Domain Knowledge

- Immediately after beginning the analysis, I prioritized visualization and consequently misunderstood the criteria for issuing ozone advisories in the U.S. I subsequently verified the threshold at 70 ppbv.
- Learned that conducting a thorough review of the domain—both before and after the analysis—is essential for ensuring the reliability of the results.

2. Systematization of the hypothesis development process

- Cultivated the habit of formulating specific hypotheses and documenting and managing them to stay focused on analysis objectives.
- Learned that clearly defining the problem leads directly to clear and necessary data analysis.

3. Ability to use tools

- Understanding the detailed features of Python and Seaborn.
- Enables the formulation of plans based on the specific capabilities of the tools used for the envisioned analysis and visualization.